Fact-Checking Pipeline for Detecting Hallucination in LLMs

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Abstract

Large Language Models (LLMs) have become widely adopted for their ability to generate human-like text and even reason through complex problems. However, they are still prone to hallucinations, correct-sounding outputs that are factually-incorrect, making them difficult to trust. Existing solutions grounds the LLMs during generation but do not verify the final responses.

This paper presents a fact-checking pipeline designed to detect hallucinations in LLM outputs and provide transparent evidence. The pipeline first condense the LLM's response into shorter, "summarized" facts that need to be checked. These claims are then used as queries for retrieval from the knowledge corpus to obtain supporting evidence. Finally, a fact-verification model detects if the ground truth supports or refutes the claim to inform the user of possible hallucinations in the LLM's response.

Keywords: LLM, Hallucination, Fact Extraction, Fact Verification, Knowledge Base,

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1 Introduction

Large Language Models (LLMs) have shown great ability to generate coherent, well-written text. Recently, reasoning models like GPT-5 and Gemini 2.5 Pro have even integrated "thinking" into their generation, performing a long chain of thought before creating a final response. These advancements have improved LLMs' ability to "think" through complex prompts to generate a seemingly correct and well-thought-out answers, encouraging more adoption of LLM use in both industry and daily life.

Despite these advancements, hallucinations, or when LLMs generate responses that sound correct but are factually-incorrect, remains a persistent and critical challenge. Research by OpenAI have shown that their reasoning models such as gpt-5-thinking and OpenAI o-mini in fact hallucinate more than the previous non-reasoning models. As a user, it is difficult to trust these responses and requires manual fact checking. Existing solutions such as Retrieval-Augmented Generation (RAG) support LLMs by providing information to base the responses on during generation, but fails to verify the final responses as factually-correct [1]. This lack of after-generation verification motivates this project to create a pipeline that combines fact extraction, evidence retrieval and fact verification to detect LLM hallucinations with supporting evidence.

The fact-checking pipeline consists four stages that take the LLM's response and outputs any hallucinated fact with evidence:

1. **Fact Extraction**: Extract atomic facts from the LLM's response
2. **Knowledge Base**: Compile trusted sources to create the knowledge base
3. **Evidence Retrieval**: Use extracted facts as queries to knowledge base and retrieve supporting evidence
4. **Fact Verification**: Classify extracted facts as *supported*, *refuted*, or *unverifiable* using retrieved evidence.

Fact extraction is implemented using Google's flan-t5 model that is fine-tuned to this specific task using the Fact Extraction and VERification (FEVER) dataset. The FEVER dataset contains individual facts extracted from Wikipedia which are manually annotated as "Supported", "Refuted" or "NotEnoughInfo". These annotation includes the support sentences from Wikipedia pages which are used as the evidence in this project. The dataset is then further processed to fit the need for fact extraction. The knowledge base is built as a vector database using trusted sources such as Wikipedia and stored as embeddings. As such sources may not have information that is recent or obscure, a web searching API updates the knowledge base with new trusted information if that information is not available. Using the extracted facts as queries, supporting evidence is retrieved from the knowledge base using ranking algorithm to find the best match. Finally,

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fact verification uses a fine-tuned NLI model that determines whether the evidence can support the corresponding fact and displays the evidence if it refutes the claim. [PLACEHOLDER FOR PIPELINE RESULTS]

This paper is organized as follows. Section 2 reviews the related work in fact verification and hallucination detection. Section 3 presents the proposed pipeline and Section 4 details the implementation. Section 5 presents the evaluation methods and results. Section 6 concludes the paper and discuss key findings and future work.

References

[1] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.